UNIT 2

1. Explain the importance of regularization in deep neural networks. Compare L1 regularization, L2 regularization, and dropout.

2. Consider the objective function $J(w) = (w-3)^2$, with derivative $dJ/dw = 2(w-3)$. Starting with $w_0 = 8$, perform three iterations of gradient descent using $\eta = 0.1$. Evaluate w1, w2, and w3. Also calculate J(w) after the third iteration.

3. Consider $J(w) = 1/2(w-6)^2$ with gradient $dJ/dw = w - 6$. Starting with $w_0 = 2$, perform four iterations of gradient descent using $\eta = 0.25$. Evaluate w_1, w_2, w_3, and w_4. Also calculate J(w4).

4. Explain the working principles of Stochastic Gradient Descent, RMSProp, and Adam. Along with explain why adaptive optimizers are useful in training deep neural networks.

5. Explain the role of learning rate in gradient-based optimization. Discuss consequences of selecting a learning rate that is too small or too large.

6. Explain batch normalization and dropout. Discuss how these techniques contribute to stable training and improved generalization.

7. Explain overfitting and underfitting in deep neural networks. Discuss the bias–variance trade-off.

8. Compare standard Gradient Descent, Momentum-based Gradient Descent, RMSProp, and Adam with respect to gradient handling and convergence behaviour.

9. A feedforward neural network consists of an input layer with 5 neurons, Hidden Layer 1 with 4 neurons, Hidden Layer 2 with 3 neurons, and an output layer with 2 neurons. All consecutive layers are fully connected and each neuron except input neurons has one bias.

(i) Illustrate the network architecture.

(ii) Evaluate the number of weights between each pair of layers.

(iii) Evaluate the total number of trainable parameters including biases.

10.For RMSProp, $v_t = \beta v_{t-1} + (1-\beta)g_t^2$ and $w_t = w_{t-1} - \eta g_t/(\sqrt{v_t} + \varepsilon)$. Given $w_0 = 2.0$, $v_0 = 0$, $\beta = 0.9$, $\eta = 0.01$, $\varepsilon \approx 0$, and gradients $g_1 = 0.8$ and $g_2 = 0.4$, calculate: (i) v1 and w1, (ii) v2 and w2.

11.A neural network parameter is updated using Adam. Given $W_0 = 1.50$, $g_1 = 0.40$, $m_0 = 0$, $v_0 = 0$, $\alpha = 0.01$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\varepsilon = 10^{-8}$

Evaluate: (i) m_1, (ii) v_1, (iii) bias-corrected estimates $\hat{m}1$ and $\hat{v}1$, and (iv) updated parameter w1. Use $m_t = \beta_1 m_{t-1} + (1-\beta_1)g_t$ and $v_t = \beta_2 v_{t-1} + (1-\beta_2)g_t^2$.

12.A parameter is optimized using Adam. At a particular iteration, $\hat{m}t = 0.24$ and $\hat{v}t = 0.09$. Current parameter $w_t = 2.50$. Use $\alpha = 0.02$ and $\varepsilon \approx 0$. Calculate: (i) parameter update magnitude, (ii) updated parameter value. Then, if gradient direction reverses so that $\hat{m}t = -0.18$ while $\hat{v}t = 0.09$, calculate the new update and comment on the direction of parameter change.